

Steering Is Cheaper Than Banning: Assistance-Based Selection of Attention’s Spectral Solutions

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Abstract

A companion study showed that the positional scheme sets the *default* spectral algebra of the attention solutions a transformer learns — rotational (RoPE) vs. content-like (learned-absolute) previous-token heads — and that this default is a training *preference*, not a hard constraint: constraints that ban a spectral channel are rerouted around at a significant, quantifiable search cost (up to $2.9\times$ slower formation). Here we ask the constructive converse: can pure *assistance* — initialization or regularization toward a chosen spectral algebra, with no bans — accelerate or *select* which functionally-equivalent implementation a circuit forms? On a controlled induction task (2-layer models, two positional schemes, $n=5$ seeds, pre-registered), we find three things. (1) **Solution-specific initialization accelerates formation** in both schemes (-28% RoPE, -37% APE; rank-biserial $+1.0$, $p=0.004$) at zero capability cost, whereas initializing merely the algebra *class* does not, and regularizing toward the default *slows* formation monotonically — assistance is a property of solution-specific seeds, not of spectral fields. (2) **Weak assistance selects the implementation**: pushed toward the non-default spectral form, 5/5 runs form the non-default implementation vs. 0/5 free, in both schemes (Fisher $p=0.004$), at full capability — and a dissection shows the weak regularizer alone is the *selector* (flipping at zero formation cost under RoPE) while initialization alone is the *accelerator*. (3) **Steering is cheaper than banning**: the same implementation flip that costs $2.9\times$ to force by constraint is bought by assistance at $1.3\times$ — and the assistance cost tracks the *target’s* intrinsic search difficulty (negative for RoPE’s near-parity alternative, $+30\%$ for APE’s hard symmetric target). We frame these as an early, toy-scale demonstration that the spectral economy of attention is not only measurable but *steerable*, and state the open mechanism (planted RoPE rotations act as scaffolds, not seeds) and scale caveats plainly.

1 Background and question

The companion paper (1) establishes, across seven pretrained models and three positional schemes, that the complex spectrum of the QK operator $M=W_q^\top W_k$ is the *fingerprint* of the algorithm a head implements: previous-token heads are spectrally rotational under RoPE and content-like under learned-absolute/ALiBi positions; the profile is consolidated *after* circuit formation; and constrained training reroutes around any banned spectral channel at a significant search cost (formation-time penalty up to $2.9\times$; all pre-registered contrasts $q_{\text{BH}}\leq 0.016$). That established a *negative* lever: banning is possible but expensive and reroutable. This note tests the *positive*

lever — assistance — and asks whether the same spectral economy that prices bans also prices, and enables, *steering*.

Concretely: (Q1) does assistance toward a scheme’s default algebra *accelerate* circuit formation? (Q2) does weak assistance toward the *non-default* algebra *select* which of two functionally-equivalent implementations forms, without bans? (Q3) how does the cost of steering compare to the cost of banning the same outcome? Predictions were frozen before runs (P-A1/P-A2/P-A3 in the program’s roadmap); we report falsified predictions with the same prominence as confirmations.

2 Setup

We reuse the companion paper’s constrained-training harness: 2-layer attention-only models ($d=128$, 4 heads, $d_k=32$) trained from scratch on a per-sequence random-map induction task (successor prediction requires finding the previous occurrence of the current token and copying), $n=5$ seeds, formation step defined as the first eval crossing the induction-CE threshold (verbatim from the companion). Free (unassisted) arms reproduce the companion’s numbers exactly (600 ± 0 APE, 940 ± 55 RoPE). Assistance comes in three forms, all *ban-free*: (i) *solution-init* — initialize the seed heads’ QK factors to carry the target implementation’s positional offset (RoPE: the $\Delta=-1$ phase θ ; APE: a symmetric nearby-position plant); (ii) *algebra-init* — initialize only the target’s algebra *class* (random phases / a skew plant) without the specific offset; (iii) *assist-reg* — a bounded complement penalty pushing the population’s mean spectral share toward the target, with detached denominators. Placebos apply the cross-scheme algebra (RoPE-style under APE and vice versa) and a random orthogonal plant. Implementation class is read out by the companion’s weight batteries (`rope_imag_frac` for RoPE, `dir_frac` for APE) with thresholds pre-set from the companion’s bimodal distributions. All outputs and per-run tables are released.

3 Results

(1) Assistance accelerates — but only as solution-specific initialization. Figure 1 and Table 1: solution-init speeds formation in both schemes (RoPE $940\rightarrow 680$, -28% ; APE $600\rightarrow 380$, -37% ; one-sided exact MWU $p=0.004$, rank-biserial $+1.0$), at *zero* capability cost (final induction CE unchanged; the solution-init arms end with *higher* previous-token scores). The pre-registered primary prediction P-A1 — that initializing the algebra *class* suffices — is **falsified as frozen**: algebra-init is non-significant in both schemes (RoPE $q_{BH}=0.16$, suggestive $rb=+0.64$; APE null at the eval-grid resolution). Placebos are null (P-A2 passes): no cross-scheme or random plant speeds formation, so the effect is target-specific, not an initialization-scale artifact. A reported surprise: *assist-reg* toward the default *slows* formation, monotonically in λ (up to $3.9\times$ at APE $\lambda=10$, with capability damage) — mirroring the companion’s constraint-cost law from the opposite sign. *Any global spectral field — toward or away from the default — has a search cost; what accelerates is a solution-specific seed, not a field.*

(2) Weak assistance selects the implementation. Pushed weakly toward the *non-default* form, every run forms the non-default implementation: 5/5 vs. 0/5 free, in *both* schemes (Fisher exact $p=0.004$ each; zero unclassified). Under RoPE-assist every final previous-token head is cos-only (`rope_imag_frac` $\in [0.016, 0.043]$, kernel peak still at $\Delta=-1$) vs. 5/5 phase-carried when free (`rope_imag_frac` ≈ 0.52); under APE-assist every head is symmetric (`dir_frac` ≈ 0.012) vs. 5/5 antisymmetric when free. Capability is unchanged (P-A3 selection clause passes maximally). A pre-registered dissection grid (108 runs) then separates the two ingredients: *the weak live regularizer is the selector and the initialization is the accelerator* — reg-only flips the implementation 5/5 in

arm	RoPE formation	APE formation	final capability
free	940±55	600±0	floor
algebra-init	840±89 (ns)	600±0 (ns)	floor
solution-init	680±45 (−28%)	380±45 (−37%)	floor (higher prev)
assist-reg $\lambda=1$	1080±45 (slower)	960±55 (slower)	floor
assist-reg $\lambda=10$	1380±130 (slower)	2340±219 (slower)	APE damaged

Table 1: Assistance-hinge formation steps ($n=5$; one-sided exact MWU vs. free, BH-FDR). Solution-specific initialization accelerates both schemes at rank-biserial +1.0, $p=0.004$; algebra-class init and regularization fields do not.

implementation flip	ban cost (companion)	assist cost (this note)	selection
→ RoPE cos-only (non-default)	+36–52%	− 23% (faster)	5/5 vs 0/5
→ APE symmetric (non-default)	2.9×	+ 30%	5/5 vs 0/5

Table 2: Steering vs. banning for the same implementation flip (Fisher $p=0.004$ each). Assistance selects the same outcome a ban forces, far more cheaply; the assist cost tracks the target’s intrinsic search difficulty.

both schemes (under RoPE at *zero* formation cost: the formation-step multiset is identical to free), while init-only accelerates without flipping (0/5 both schemes); the combined arm’s cost decomposes accordingly (≈ 0 RoPE, +30% APE, carried entirely by the selection).

(3) Steering is cheaper than banning. The cost of assisted selection tracks the *target’s* intrinsic search difficulty, and is far below the ban cost for the same outcome. Forming RoPE’s cos-only implementation costs −23% under assistance (*faster* than free) vs. +36–52% to force it by banning the phase channel (companion); forming APE’s symmetric implementation costs +30% under assistance vs. 2.9× to force it by banning antisymmetry. *Assisted selection* $\ll$ *constrained rerouting for the same implementation flip* (Table 2). The two levers also have different cost *geometries*: a λ -grid over 33× shows the ban’s cost is a *plateau* — the constraint binds already at $\lambda=0.3$ and the delay is λ -insensitive thereafter (the cost *is* the reroute, a step function of whether the ban binds; an earlier apparent non-monotonicity did not replicate at $n=5$ and is retired) — whereas toward-default field pressure grows monotonically and, under APE, superlinearly with capability damage.

4 Open mechanism and caveats

Scaffold, not seed. Under RoPE the planted solution is *never* adopted — in all 5/5 acceleration runs the planted heads end with $\text{prev} \approx 0.02$ and the circuit forms in *unseeded* heads — yet formation still accelerates 28%. The RoPE plant acts as a *scaffold* (candidate mechanisms: gradient shaping through the second layer / OV, or the plant’s coherent structure aiding early representation formation), not a seed. Under APE the plant *is* adopted (5/5). A pre-registered freeze test sharpens the scaffold reading: *freezing the plant entirely* (no gradient into the seeded heads) preserves — in fact *doubles* — the speedup (500 median vs. 700 unfrozen vs. 900 free; $\text{rb} = +1.0$, capability intact). The seed need not even be trainable; training’s own modification of the planted heads apparently *degrades* the scaffold, pointing at a representational/gradient-shaping mechanism rather than plant adaptation — which we still flag as open rather than adjudicated. **Caveats:** $n=5$; 2-layer attention-only models on one synthetic task; APE formation is quantized at the 100-step

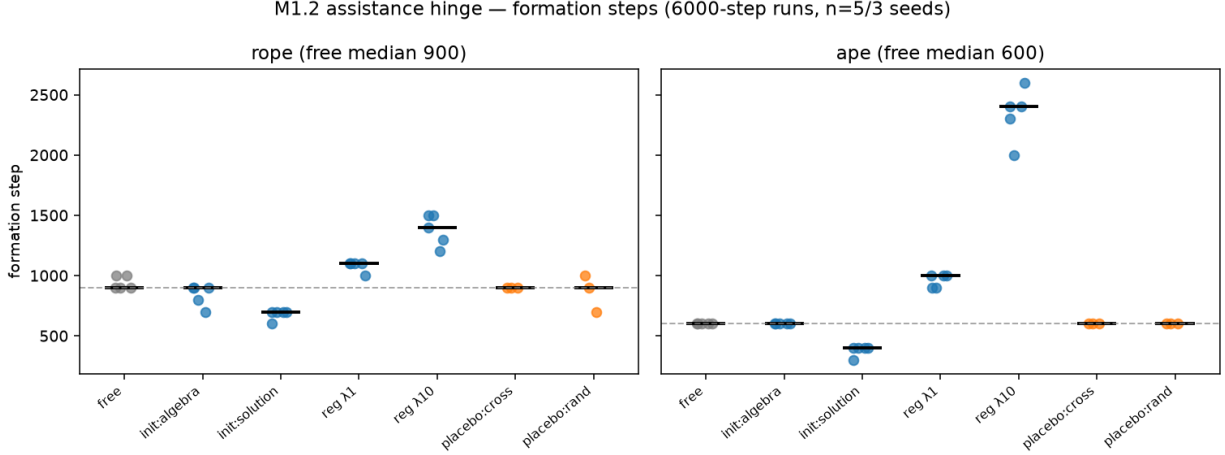


Figure 1: Formation steps by assistance arm (left RoPE, right APE; $n=5/3$ seeds). Solution-init (third column) accelerates both schemes; algebra-init and placebos sit at free; regularization fields slow formation, monotonically in λ .

eval grid (so the APE algebra-init null means “no effect at 100-step resolution,” not a measured zero). These results are an existence demonstration of steerability at toy scale, not a scaling law; the program’s next milestone tests assisted selection at $\geq 160\text{M}$ -parameter pretraining through the formation window.

5 Takeaway

The spectral economy of attention is not only measurable and priced (companion paper) but *steerable*: weak, ban-free assistance selects which functionally-equivalent spectral implementation a circuit adopts, at a cost far below banning the alternative, and solution-specific initialization accelerates circuit formation outright. If this holds at scale, “which algorithm a circuit implements” becomes a controllable design choice — relevant to interpretability-by-construction and to installing chosen (e.g., more auditable) implementations. Steering is also *auditable* where rerouting is not hideable: companion propagator-level work finds that implementations which vanish from the weight fingerprint (`rope_imag_frac` ≈ 0.003 after a phase ban) remain fully visible as transient-reserve excess on the attention propagator (31/31 prev-carrying heads positive across all implementations) — the weight battery reads *implementation*, the propagator battery reads *function*, so a steered or rerouted circuit cannot evade both. We release the harness, seeds, and per-run tables.

Reproducibility & pre-registration. Harness `src/{h1_hinge,h1_analyze,h2_m21}.py` (extends the companion’s p10 additively; free arms bit-exact); predictions P-A1/P-A2/P-A3 and the follow-up family P-M21a–d/P-A5/P-E2 frozen before their respective runs; per-run tables `results/h1/m12_summary.parquet`, `m13_summary.parquet`, `results/h2/`; falsified predictions (P-A1; the λ non-monotonicity flag P-M21c) reported as findings.

References

- [1] Hengyu Li. Fingerprint, not blueprint: How positional schemes set the default spectral algebra of attention, 2026. Companion paper; preprint.